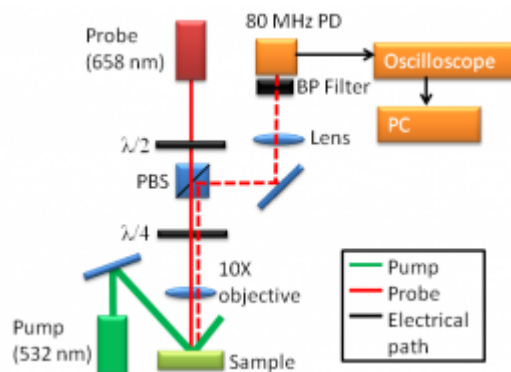


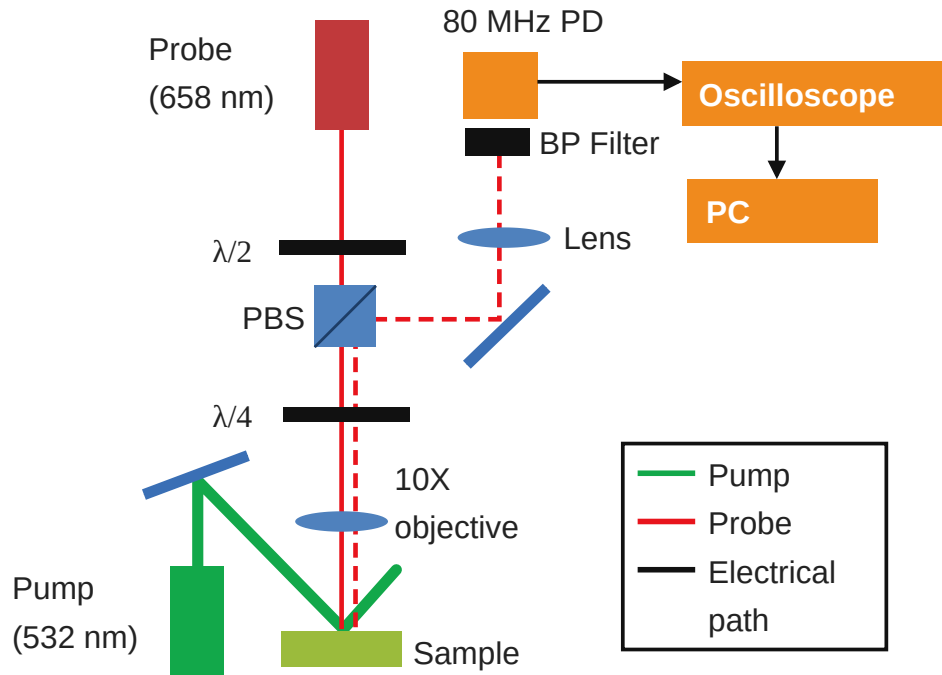
Nanosecond Transient Thermoreflectance

Source: <https://nanoheat.stanford.edu/tools/nanosecond-transient-thermoreflectance/> · Stanford NanoHeat Lab (Prof. K. E. Goodson), Stanford University

The nanosecond thermoreflectance technique (sometimes called transient thermoreflectance, TTR) has been a fixture in our laboratory for decades. This method characterizes the thermal properties of sample structures by optically measuring the transient thermal response to nanosecond heating pulses. The technique is particularly useful for measuring the thermal properties and boundary resistances of thin films ranging in thickness from hundreds of nanometers up to microns. The technique uses 6 ns pulses from a neodymium-doped yttrium aluminum garnet (Nd:YAG) laser operating at 10 Hz to heat the top metal film with a spot diameter of ~ 3 mm, creating a transient temperature field in the sample. The reflectivity of the metal film is proportional to the surface temperature and is measured through the reflected intensity of a continuous wave probe laser. Because the heating diameter is much larger than the maximum thermal diffusion depth, data interpretation can rely on solutions of the one-dimensional heat diffusion equation for a multilayer stack. A multi-parameter least-squares fit of the transient thermal response data to the solution of the thermal model extracts the effective thermal properties of the film and its interfaces. The confining of heat to the film due to the short timescales, and unique time-dependent sensitivity of the thermal response to the thermal properties of the film enable profiling of thermal resistances within the film and at its interfaces from a single thermal trace.



원본 그림 (267×191 px, 페이지에 실린 크기 그대로). Nanosecond transient thermoreflectance 광학계.



[제작도] 원본이 저해상도라 같은 구성을 벡터로 다시 그린 도면. 구성 요소·라벨·경로는 원본과 동일하게 옮겼으며, 위치와 비율은 근사치임.

광학계 구성 (그림에서 읽은 내용)

- **Pump (532 nm):** 미러로 꺾어 시료 표면에 비스듬히 입사 (녹색 경로).
- **Probe (658 nm):** $\lambda/2$ 판 → PBS → $\lambda/4$ 판 → 10X 대물렌즈를 거쳐 시료에 수직 입사 (적색 실선).
- **반사 probe:** 같은 대물렌즈· $\lambda/4$ 판을 되돌아와 PBS에서 분리 → 미러 → 렌즈 → BP(밴드패스) 필터 → 80 MHz 포토디텍터 (적색 점선).
- **신호 경로:** PD → 오실로스코프 → PC (흑색, electrical path).

※ 본문에는 펌프가 Nd:YAG 6 ns / 10 Hz, 스폿 지름 ~3 mm로 적혀 있고 파장은 명시되어 있지 않음. 그림의 532 nm는 Nd:YAG 2차 고조파에 해당. 그림의 10X 대물렌즈는 probe 집속용으로 보임.